# Project Mail: $M/G/3/3$ Loss System with Weather-Dependent Service Times

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# Setup

- Messages arrive according to a Poisson Process with rate

$$\lambda = 2 \text{ per hour.}$$

- There are 3 channels. If at least one channel is free, the message gets serviced immediately.
- If all three channels are busy, the message is lost. No queuing or wait time.
- Service time depends on the weather at the moment of arrival.
- Weather-dependent service distributions:

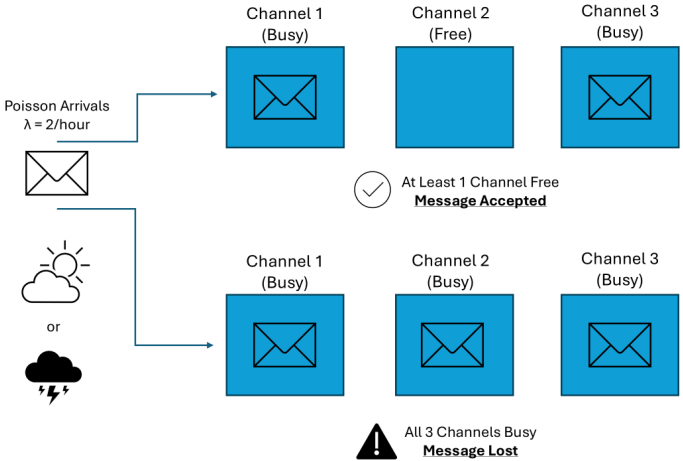
$$F_G(x) = x, \qquad 0 < x < 1,$$

$$F_B(x) = x^3, \qquad 0 < x < 1.$$

- Goal is to estimate the expected number of messages lost in the first 100 hours.

# System Structure

M/G/3/3 Loss System with Weather-Dependent Service Times




Weather States: Bernoulli($p$)

Bad Weather with prob. 1-$p$
Beta(3,1)
$F(x) = x^3$, $0 < x < 1$

Good Weather with prob. $p$
Uniform(0,1)
$F(x) = x$, $0 < x < 1$

Model:

- Message Arrives
- Realize Weather State
- Assign Service Time
- Enter Free Channel or Message Lost

# Assumptions

- Sequence of events:

$$\text{arrival} \to \text{weather realization} \to \text{service time draw} \to \text{service or lose.}$$

- Arrival process:

$$M(t) \sim \mathrm{Poisson}(2t).$$

- Weather state at each arrival is Bernoulli:

$$W \sim \mathrm{Bernoulli}(p),$$

  where

$$\mathbb{P}(W = 1) = p, \qquad \mathbb{P}(W = 0) = 1 - p.$$

- $W = 1$: good weather, $W = 0$: bad weather.
- Conditional service time distributions:

$$X \mid W = 1 \sim \mathrm{Uniform}(0, 1), \qquad X \mid W = 0 \sim \mathrm{Beta}(3, 1).$$

# Service Time Distribution

Conditioning on the weather state,

$$X = X_G \mathbf{1}_{\{W=1\}} + X_B \mathbf{1}_{\{W=0\}}.$$

Thus the service time distribution is a two component mixture:

$$F_X(x) = pF_G(x) + (1-p)F_B(x), \qquad 0 < x < 1.$$

Substituting,

$$F_X(x) = px + (1-p)x^3, \qquad 0 < x < 1.$$

Differentiating gives the PDF,

$$f_X(x) = p + 3(1-p)x^2, \qquad 0 < x < 1.$$

This mixture is the only place where weather enters the loss probability calculation.

# Expected Service Time

By the law of total expectation,

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid W]\big] = p\,\mathbb{E}[X \mid W = 1] + (1-p)\,\mathbb{E}[X \mid W = 0].$$

The component first moments are

$$\mathbb{E}[X \mid W = 1] = \frac{1}{2}, \qquad \mathbb{E}[X \mid W = 0] = \frac{3}{4}.$$

Thus

$$\boxed{\mathbb{E}[X] = \frac{3-p}{4}}.$$

Similarly,

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{3}{80} + \frac{p}{120} - \frac{p^2}{16}}.$$

# Erlang Insensitivity

For an $M/G/C/C$ loss system, the stationary blocking probability is *insensitive* to the shape of the service distribution and depends only on the mean service time.

Thus the $M/G/3/3$ system can be analyzed using the Erlang-B formula. Let

$$a = \lambda \mathbb{E}[X].$$

Using

$$\lambda = 2, \qquad \mathbb{E}[X] = \frac{3-p}{4},$$

the offered load (effective demand on the system) is

$$\boxed{a = 2 \cdot \frac{3-p}{4} = \frac{3-p}{2}}.$$

This simplifies the problem considerably.

# Loss Probability (Erlang-B)

Let $p_k$ denote the steady-state probability of $k$ busy channels. For an $M/G/3/3$ loss system,

$$p_k = \frac{a^k/k!}{\sum_{j=0}^{3} a^j/j!}, \qquad k = 0, 1, 2, 3,$$

with offered load $a = (3-p)/2$. A message is lost exactly when all 3 channels are busy, thus

$$\mathbb{P}(\text{message lost}) = p_3.$$

Using the Erlang-B formula,

$$p_3 = \frac{a^3/3!}{1 + a + a^2/2! + a^3/3!} = \frac{a^3/6}{1 + a + a^2/2 + a^3/6}.$$

Substituting $a = (3-p)/2$ gives

$$\boxed{p_3 = \frac{(3-p)^3}{201 - 87p + 15p^2 - p^3}.}$$

# Expected Number of Messages Lost

Let $L_m(t)$ be the number of lost messages in the interval $[0, t]$.

Since the arrival process has rate $\lambda = 2$ and each arrival is blocked with probability $p_3$,

$$\mathbb{E}[L_m(t)] = \lambda t \cdot p_3 = 2t \cdot p_3.$$

Substituting and using $t = 100$,

$$\boxed{\mathbb{E}[L_m(t)] = 200 \left[ \frac{(3-p)^3}{201 - 87p + 15p^2 - p^3} \right].}$$

I use this to compare to simulation results.

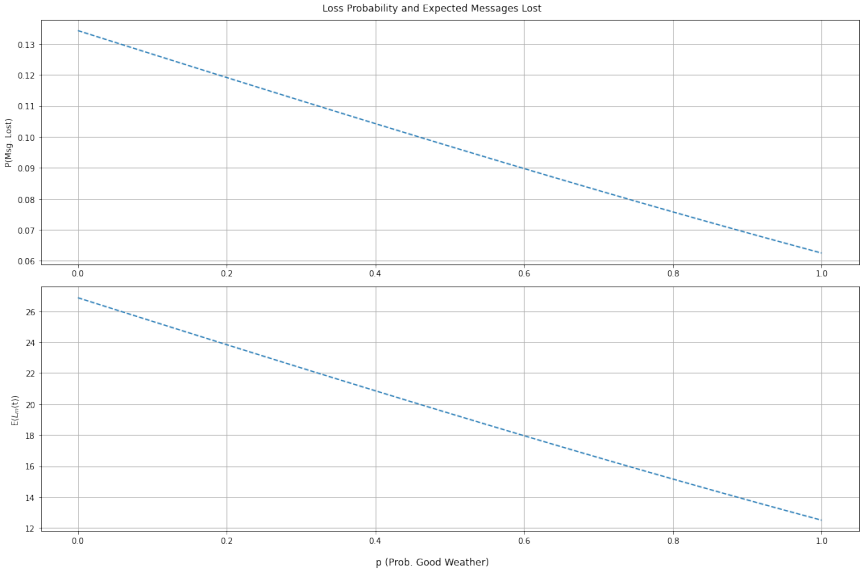
# Three Benchmark Cases

| Case | $\mathbb{E}[X]$ | $p_3$ | $\mathbb{E}[L_m(100)]$ |
|---|---|---|---|
| Always bad weather ($p = 0$) | $\frac{6}{8}$ | $\approx 0.134328$ | $200(0.134328) \approx 26.87$ |
| Equal mixture ($p = 1/2$) | $\frac{5}{8}$ | $\approx 0.096974$ | $200(0.096974) \approx 19.39$ |
| Always good weather ($p = 1$) | $\frac{4}{8}$ | $0.0625$ | $200 \cdot \frac{1}{16} = 12.5$ |

As $p$ increases (good weather more likely)

- Expected service time falls
- Loss probability falls
- Expected number of messages lost falls

# Loss Probability and Expected Messages Lost Across $p$



# Simulation Exercise

To validate the analytic results, I simulate the process over $[0, 100]$.

1. Generate arrivals from PP(2) using exponential draws for interarrival times:

$$T_{i+1} - T_i \sim \mathrm{Exp}(2).$$

2. At each arrival time $T_i$, remove all messages that have been processed.
3. Draw weather state

$$W_i \sim \mathrm{Bernoulli}(p).$$

4. Draw service time

$$X_i = \begin{cases} U_i, & W_i = 1, \\ U_i^{1/3}, & W_i = 0, \end{cases} \qquad U_i \sim \mathrm{Uniform}(0, 1).$$

5. If fewer than 3 channels are busy, process the message and schedule departure at $T_i + X_i$.
6. If all 3 channels are busy, record a lost message.

# Simulation Outputs

For each simulated path, the main statistics are the count of messages lost,

$$L_m(t) = \sum_{i=1}^{M(t)} L_i, \qquad L_i = \mathbf{1}\{\text{arrival } i \text{ is lost}\},$$

the fraction of messages lost,

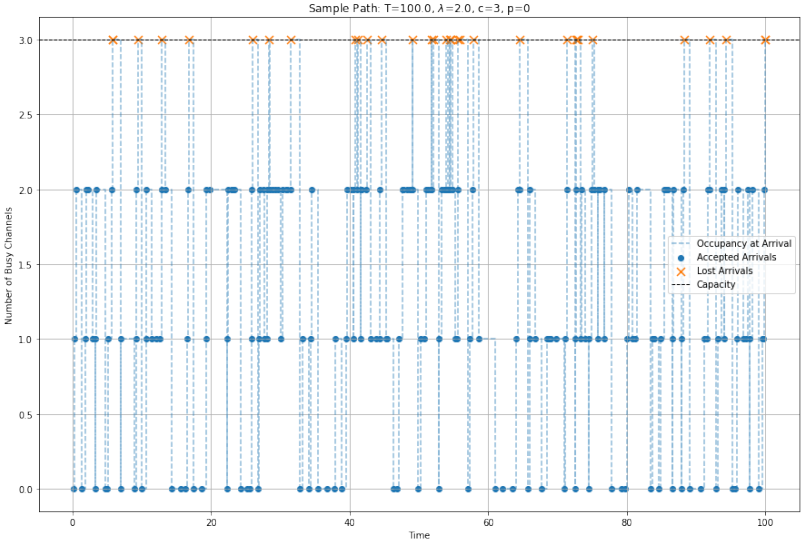
$$\widehat{p}_3 = \frac{1}{M(t)} \sum_{i=1}^{M(t)} L_i,$$

and the empirical mean service time over accepted jobs,

$$\widehat{\mathbb{E}[X]} = \frac{1}{N_{\text{accepted}}} \sum_{i=1}^{N_{\text{accepted}}} X_i.$$

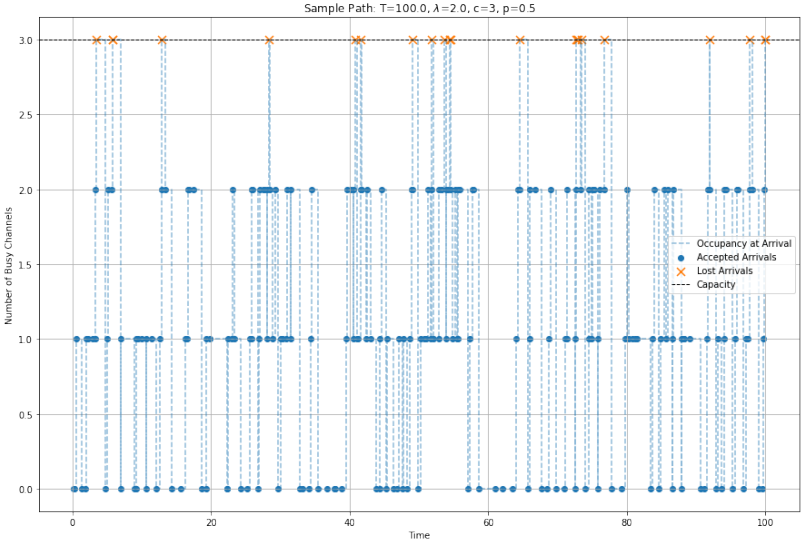
I compare the sampling distributions of the empirical quantities generated from $N = 1000$ runs against $\mathbb{E}[X]$, $p_3$, and $\mathbb{E}[L_m(100)]$.

# Sample Path: Always Bad Weather ($p = 0$)



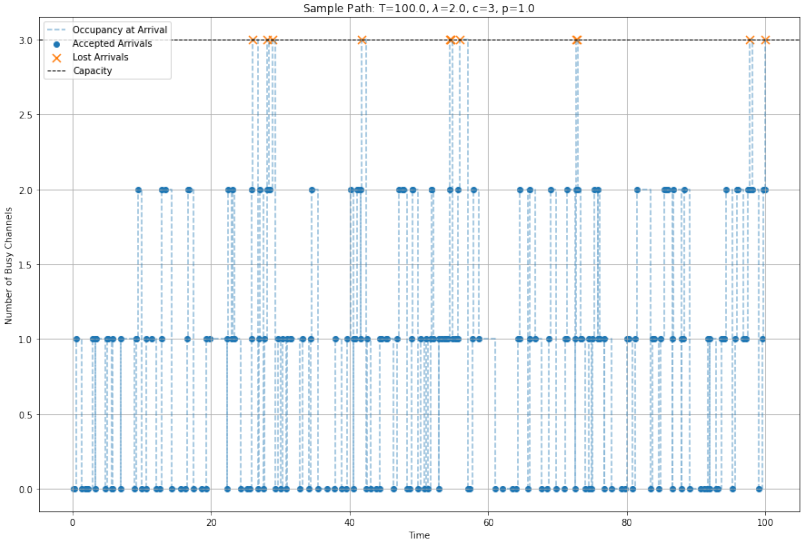

- Longer service times result in more congestion.
- System frequently operates near capacity ($c = 3$).
- Losses cluster during sustained busy periods.

# Sample Path: Mixed Weather ($p = 0.5$)




- Shorter periods of sustained full capacity.
- Losses less frequent than in the $p = 0$ case.
- Higher probability of good weather results in lower offered load.

# Sample Path: Always Good Weather ($p = 1$)




- Service times are shortest on average.
- System rarely remains at sustained full capacity.
- Message loss is isolated (minimal clustering) and infrequent.

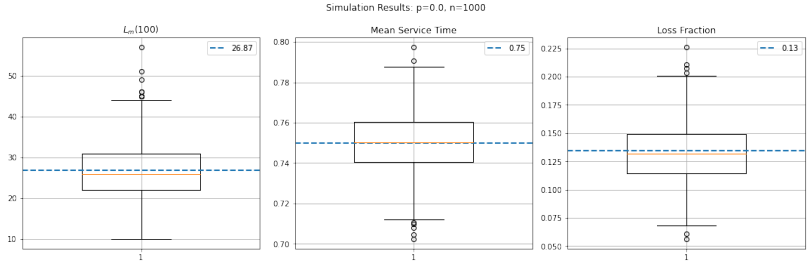
# Monte Carlo Results

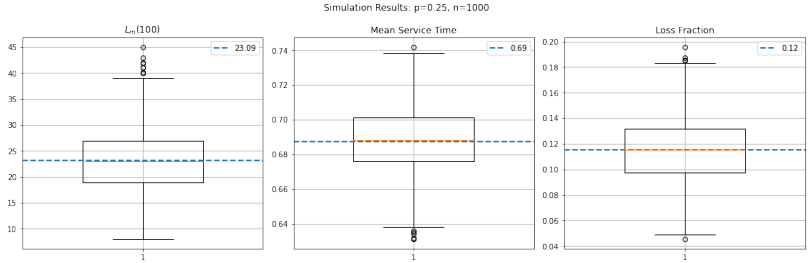
**Monte Carlo Estimates vs. Theoretical Benchmarks**

| | Number Lost | | | Service Time | | | Loss Fraction | | |
|---|---|---|---|---|---|---|---|---|---|
| $p$ | Mean | Std. | $\mathbb{E}[L_m(100)]$ | Mean | Std. | $\mathbb{E}[X]$ | Mean | Std. | $p_3$ |
| 0.00 | 26.69 | 6.77 | 26.87 | 0.750 | 0.015 | 0.750 | 0.132 | 0.027 | 0.134 |
| 0.25 | 23.18 | 6.07 | 23.09 | 0.688 | 0.019 | 0.688 | 0.115 | 0.025 | 0.115 |
| 0.50 | 19.79 | 5.58 | 19.39 | 0.626 | 0.021 | 0.625 | 0.098 | 0.023 | 0.097 |
| 0.75 | 15.72 | 5.01 | 15.85 | 0.563 | 0.021 | 0.563 | 0.078 | 0.022 | 0.079 |
| 1.00 | 12.55 | 4.50 | 12.50 | 0.500 | 0.022 | 0.500 | 0.062 | 0.020 | 0.063 |

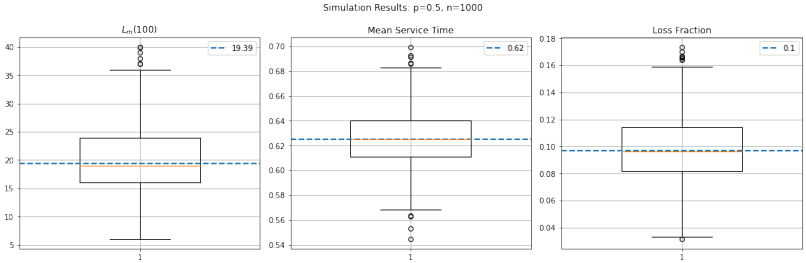
- Sampling distributions constructed from 1000 independent runs per value of $p$.
- Empirical means closely match theoretical predictions across all metrics.
- Variability decreases as $p$ increases (lower offered load).

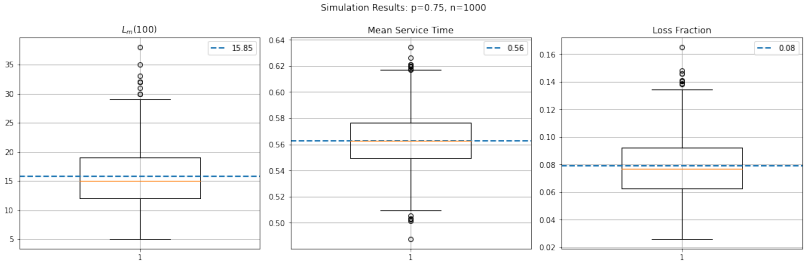
# Monte Carlo Validation: $p = 0$ and $p = 0.25$



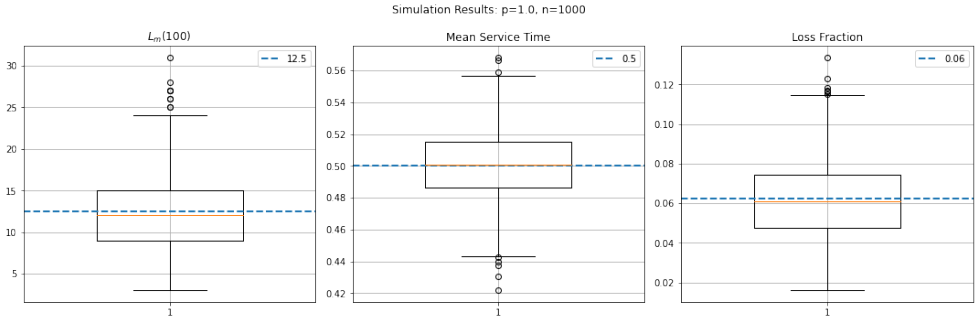



# Monte Carlo Validation: $p = 0.5$ and $p = 0.75$



# Monte Carlo Validation: $p = 1$




- MC sampling distributions are tightly centered at theoretical values.
- Mean service time simulation results align with analytic values.
- Variability is lowest in the always good weather case ($p = 1$).
- Positive skewness in distribution of number of messages lost and loss fraction (congestion events).

# Summary

- Model as an $M/G/3/3$ loss system with weather-dependent service times.
- The service time distribution is a Bernoulli mixture with
$$\mathbb{E}[X] = \frac{3-p}{4}.$$
- By the insensitivity property, message loss depends only on the mean service time, so Erlang-B applies with
$$a = \frac{3-p}{2}.$$
- The probability of a lost message is
$$p_3 = \frac{(3-p)^3}{201 - 87p + 15p^2 - p^3},$$
and the expected number of lost messages in 100 hours is
$$\boxed{\mathbb{E}[L_m(100)] = 200\, p_3}.$$
- Simulation results are consistent with the analytic predictions across tested $p$ values.

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# Appendix: Derivation of $\mathbb{E}[X]$

By the law of iterated expectation,

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid W]\big] = p\,\mathbb{E}[X \mid W = 1] + (1-p)\,\mathbb{E}[X \mid W = 0].$$

Using

$$\mathbb{E}[X \mid W = 1] = \int_0^1 x\,dx = \frac{1}{2},$$

and

$$\mathbb{E}[X \mid W = 0] = \int_0^1 x\,(3x^2)\,dx = 3\int_0^1 x^3\,dx = \frac{3}{4},$$

I obtain

$$\mathbb{E}[X] = p \cdot \frac{1}{2} + (1-p) \cdot \frac{3}{4} = \frac{2p + 3 - 3p}{4} = \boxed{\frac{3-p}{4}}.$$

# Appendix: Derivation of $\mathrm{Var}(X)$

First compute

$$\mathbb{E}[X^2] = p\,\mathbb{E}[X^2 \mid W = 1] + (1-p)\,\mathbb{E}[X^2 \mid W = 0].$$

Now

$$\mathbb{E}[X^2 \mid W = 1] = \int_0^1 x^2\,dx = \frac{1}{3}, \qquad \mathbb{E}[X^2 \mid W = 0] = \int_0^1 x^2(3x^2)\,dx = \frac{3}{5}.$$

Thus

$$\mathbb{E}[X^2] = p \cdot \frac{1}{3} + (1-p) \cdot \frac{3}{5} = \frac{9-4p}{15},$$

and

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{9-4p}{15} - \left(\frac{3-p}{4}\right)^2 = \boxed{\frac{3}{80} + \frac{p}{120} - \frac{p^2}{16}}.$$

# Appendix: Computing $p_3$

Using

$$p_3 = \frac{a^3/6}{1 + a + a^2/2 + a^3/6}, \qquad a = \frac{3-p}{2},$$

and substituting $a$ gives

$$\begin{aligned}
p_3 &= \frac{(3-p)^3/48}{1 + (3-p)/2 + (3-p)^2/8 + (3-p)^3/48} \\
&= \frac{(3-p)^3}{48 + 24(3-p) + 6(3-p)^2 + (3-p)^3} \\
&= \frac{(3-p)^3}{48 + 72 - 24p + 54 - 36p + 6p^2 + 27 - 27p + 9p^2 - p^3} \\
&= \boxed{\frac{(3-p)^3}{201 - 87p + 15p^2 - p^3}}.
\end{aligned}$$

# Appendix: Full Monte Carlo Summary

**Number Lost**

| $p$ | Count | Mean | Std. | Min | 25% | 50% | 75% | Max | $\mathbb{E}[L_m(100)]$ |
|---|---|---|---|---|---|---|---|---|---|
| 0.00 | 1000 | 26.69 | 6.77 | 10 | 22 | 26 | 31 | 57 | 26.87 |
| 0.25 | 1000 | 23.18 | 6.07 | 8 | 19 | 23 | 27 | 45 | 23.09 |
| 0.50 | 1000 | 19.79 | 5.58 | 6 | 16 | 19 | 24 | 40 | 19.39 |
| 0.75 | 1000 | 15.72 | 5.01 | 5 | 12 | 15 | 19 | 38 | 15.85 |
| 1.00 | 1000 | 12.55 | 4.50 | 3 | 9 | 12 | 15 | 31 | 12.50 |

**Service Time**

| $p$ | Count | Mean | Std. | Min | 25% | 50% | 75% | Max | $\mathbb{E}[X]$ |
|---|---|---|---|---|---|---|---|---|---|
| 0.00 | 1000 | 0.750 | 0.015 | 0.702 | 0.741 | 0.750 | 0.760 | 0.797 | 0.750 |
| 0.25 | 1000 | 0.688 | 0.019 | 0.631 | 0.676 | 0.688 | 0.701 | 0.742 | 0.688 |
| 0.50 | 1000 | 0.626 | 0.021 | 0.545 | 0.611 | 0.625 | 0.640 | 0.699 | 0.625 |
| 0.75 | 1000 | 0.563 | 0.021 | 0.487 | 0.550 | 0.563 | 0.577 | 0.634 | 0.563 |
| 1.00 | 1000 | 0.500 | 0.022 | 0.422 | 0.486 | 0.501 | 0.515 | 0.568 | 0.500 |

**Loss Fraction**

| $p$ | Count | Mean | Std. | Min | 25% | 50% | 75% | Max | $p_3$ |
|---|---|---|---|---|---|---|---|---|---|
| 0.00 | 1000 | 0.132 | 0.027 | 0.056 | 0.115 | 0.132 | 0.150 | 0.226 | 0.134 |
| 0.25 | 1000 | 0.115 | 0.025 | 0.045 | 0.097 | 0.115 | 0.132 | 0.196 | 0.115 |
| 0.50 | 1000 | 0.098 | 0.023 | 0.031 | 0.082 | 0.096 | 0.114 | 0.174 | 0.097 |
| 0.75 | 1000 | 0.078 | 0.022 | 0.026 | 0.063 | 0.077 | 0.092 | 0.165 | 0.079 |
| 1.00 | 1000 | 0.062 | 0.020 | 0.016 | 0.048 | 0.061 | 0.075 | 0.134 | 0.063 |